M380 - HW9 - 10

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Problem 26

You can find some cipher text to decrypt on the Canvas assignment page as well as here. I included both in hopes that you can copy and paste. Assume a standard 26 character alphabet. Assume that the cipher text was encrypted with an affine cipher. Look on the Canvas Assignment for an extra SageMath file that may be helpful. Please hand in the decrypted text along with the work you did to determine the affine cipher.
Cipher Text = "lkdvgbdjkbnlculxekhucdbnjkeluxjylgdgrwzluxlukejchwzrucjwke-jnbbuglxektbrdjjhdlxekkehkhgn bdkdkbidtbrwejhwktbrkwtkbdvwjhnorkkjwwb-wkhzjdkejdbrucojqbwjtbrnhzjlktbrdkhwkkbqwjjajhdebwwbwgbb zdtbrwlxekojk-fjjukejjtjdtbrwjihwhgtajcvhrdjkeldldkewlggjwkewlggjwulxekhucubbujdxbuuhdhyjtbrqwbnkejojh dkhobrkkbdkwlzjtbrzubflkdkewlggjwkewlggjwulxektbrwjqlxekluxqbwtbrwglqjludlcjhzlggjwkewlggjwkbulxektjh edbbukejwjflggoj"
Frequency Analysys: J-46, K-44

$$
\begin{aligned}
&(9,4)\;\; e \to j \quad 4 = 9k + s \quad I \\
&(10,19)\; t \to k \quad 19 = 10k + s \quad II \\
&II - I = 15 = k \\
&s = 19 = 10(15) = s = 25 \\
&M = 1/15(c - 25)
\end{aligned}
$$

Which gives the lyrics to Thriller when plugged into sage

Problem 27

Assuming a standard 26 character alphabet, consider the result of analyzing the frequency of the appearance of letters in some cipher text. Suppose the analysis comes back that the letter D occurs 38 times, S occurs 37 times, and T occurs 36 times.
(a) Assuming that the cipher text was encrypted with an affine cipher. Make your best guess as to a formula to decrypt the cipher text.

$$
\begin{aligned}
&(3,4)\ \ e \to j \quad 4 = 3k + s \quad I \\
&(18,19)\ t \to k \quad 19 = 18k + s \quad II \\
&II - I = 15 = 15k \\
&k = 1 \\
&4 = 3(1) + s \\
&s = 1 \\
&m = 1(c-1)
\end{aligned}
$$

(b) Try your formula to decrypt the cipher TQEKM. Did it work?

Which gives "urfln" meaning our decryption is incorrect.
(c) Obviously your association from 27a was incorrect. The frequencies are so close that is to be expected. I wrote this assuming that e,t, and n occur most in the English language, devise a plan on how to crack the cipher text TQEKM and state the intended message. Hint: It may be useful to work with six friends Try all combos:

$$
\begin{aligned}
&E \to S \\
&T \to D \\
&(18,4) \quad = 4 = 18k + s \quad I \\
&(3,19) \quad = 19 = 18k + s \quad II \\
&II - I = 15 = -15k = 25 \\
&k = 25 \\
&19 = 3(25) + S = 22 \\
&s = 22
\end{aligned}
$$

When plugging into sage we get: dgsmk, which is garbage.

$$
\begin{aligned}
&E \to D \\
&T \to T \\
&(3,4) \quad = 4 = 3k + s \quad I \\
&(19,19) \quad = 19 = 19k + s \quad II \\
&II - I = 15 = -16k = 15/16 \mod 26 = DNE
\end{aligned}
$$

So that can't work either

$$
\begin{aligned}
&E \to S \\
&T \to T \\
&(18,4) \quad = 4 = 18k + s \quad I \\
&(19,19) \quad = 19 = 19k + s \quad II \\
&II - I = 15 = 1k = 15 \\
&k = 15 \\
&4 = 18(15) + S = 20 \\
&s = 20
\end{aligned}
$$

Which gives the message tacos!

Problem 28

Note: I programmed a sage math tool that takes in a message and a starting matrix and determines the encrytped message, decrypted message, as well as the inverse of the starting matrix. I can show this if more work is needed to be shown

Use the block cipher below to encrypt the message (omit spaces) m =“do not be a”

$$\begin{bmatrix} 3 & 14 \\ 13 & 14 \\ 19 & 1 \end{bmatrix} \times \begin{bmatrix} 12 & 11 \\ 13 & 7 \end{bmatrix} = \begin{bmatrix} 10 & 1 \\ 0 & 7 \\ 7 & 8 \\ 22 & 18 \end{bmatrix}$$

Which gives the C = kbahhiws

Problem 29

Assuming that the block cipher below was used for encryption, decrypt the cipher message “IWVLWE”

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & 25 \\ 21 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 22 \\ 21 & 1 \\ 22 & 4 \end{bmatrix} \times \begin{bmatrix} 3 & 25 \\ 21 & 2 \end{bmatrix} = \begin{bmatrix} 18 & 10 \\ 8 & 1 \\ 20 & 12 \end{bmatrix}$$

Which translates to ”Skibum”

Problem 30

Qualities of good encryption include a large number of possible keys along with diffusion. Consider the block cipher defined by a 3x3 grid

(a) Demonstrate diffusion by encrypting the message m1 =“muskat” and the message m2 =“muscat”. How many characters in the cipher text change when you change one character in the message?

The first is me, the second is a variety of grape

muskat - qgc**fex**

muscat - qgc**fwj**

The letters after the changed letter in M change C.

(b) Provide an upper bound to the number of possible keys for encryption with a 3 × 3 matrix.

$26^9 \approx 5.42 \times 10^{12}$ which would be very difficult to brute force

(c) Utilize technology to determine (matrix)

SageMath is recommended, but a hand held calculator can do with some additional

modular work

$$\begin{bmatrix} 5 & 23 & 24 \\ 11 & 14 & 21 \\ 11 & 5 & 1 \end{bmatrix}$$

(d) Decrypt the c = 21 20 12 00 13 01 10 07 16 15 10 01. I saved you a step by providing the numeric representation of the cipher. Each two letters corresponds to a letter

Using my sage math tool I get ”Pray for Snow”

Problem 31

The bound you most likely found in 30b is likely much larger that the actual number of 3×3 invertible matrices. Let's test this thought, write down a random 3 × 3 matrix A. Determine whether or not you can calculate the inverse A-1, my guess is you pick an invertible one approximately 20 percent of the time. Still 1/5th of the bound you found in 30b is a lot
Random Matrix:

$$\begin{bmatrix} 1 & 2 & 5 \\ 6 & 8 & 14 \\ 11 & 1 & 4 \end{bmatrix}^{-1} \quad \bmod 26 = DNE$$

Problem 32

Out of curiosity I decided to attempt a bruteforece. I tried all 2x2 matrix combinations and saved the results to a spreadsheet.

Some stats: Successful attempts: 126942 Misses due to non-invertible matrices: 263683 In 166.35 seconds

And the top words found: jpiratex
heateror
dashspur
pigbodal
pugnorab

Obviously none of these are correct, and shows another downside to bruteforce. I'd like to know if you have a solution to filter down possible matches